

Mitigating cross-modal interference in multimodal fake news detection via adaptive feature optimization

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ABSTRACT

This paper introduces Adaptive Feature Matching Optimization (AFMO) for multimodal fake news detection. AFMO integrates modality-specific feature extraction with dual outlier filtering and a simulated-annealing feature-selection module to optimize the fusion of heterogeneous signals. Its main contribution is an adaptive selection mechanism that suppresses cross-modal interference, thereby improving discrimination between authentic and fake content. Experiments on Weibo, GossipCop, and PolitiFact show that AFMO consistently outperforms recent competitive baselines. On Weibo, AFMO achieves 92.23% accuracy, supporting the claim that mitigating cross-modal interference can materially improve veracity prediction.

1. Introduction

The growing intersection between Artificial Intelligence (AI) and Operations Research (OR) represents a promising avenue for advancing intelligent decision-making and optimization (De Bock et al., 2024). AI techniques, particularly deep learning, have demonstrated remarkable capabilities in extracting rich representations from unstructured modalities such as text, images, and speech. However, they often suffer from limited interpretability and degraded robustness when confronted with noisy or ambiguous inputs. In contrast, OR provides a set of mathematically grounded methods that excel in structured decision-making and logical inference. Therefore, this paper aims to develop an optimization-aware AI framework that applies the strengths of both paradigms. Specifically, we propose an integrated approach that combines deep multimodal feature learning with principled optimization strategies. The objective is to enable interpretable and accurate detection of misinformation in complex multimodal contexts.

In fact, assuring the precision of multimodal fake news detection necessitates sophisticated techniques adept at managing the innate noise, ambiguity, and inconsistencies that frequently arise when dealing with multiple data types (Singhal et al., 2019). For example, as shown in Fig. 1, the original model failed to detect the fake news example “Goodbye Nokia to the former king of tribute July 20, which is today Nokia officially into Microsoft an era of memory ended Nokia’s current ceo Yoma Ollila announced at a press conference agreed to Microsoft acquisition when the last sentence said we did not do anything wrong but somehow we After losing dozens of Nokia executives could not help but cry which Nokia you have owned? CCTV news.” However, this news is indeed a fake news as the CEO of Nokia at that time was not Yoma Ollila, and Nokia still existed as a division of Microsoft. The image in this example is partly a Nokia cell phone. Looking at the image alone, it is not a piece of false information, so it may mask more important textual information, leading to an inadequate understanding of the news as a whole and resulting in the news being categorized as true by a number of methods before the simulated annealing algorithm is used. After incorporating the simulated annealing algorithm, the

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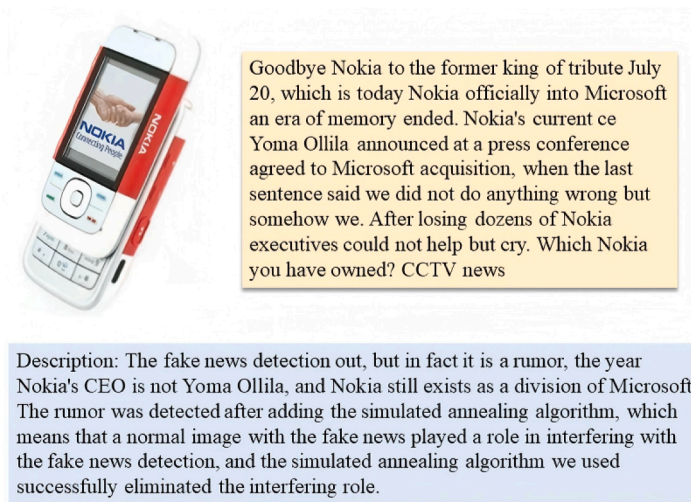


Fig. 1. The fake news is accompanied by a picture of Nokia.

fake news was successfully detected, indicating that the normal image accompanying the fake news had interfered with the detection process. Our implementation of the simulated annealing algorithm effectively eliminated this interference.

When traditional deep learning methods apply feature engineering to integrate text and image features, they commonly encounter feature interference. The primary challenges are as follows.

- **Inherent Differences:** Text and images are inherently different data types. Text is discrete, serialized, and rich in semantics, whereas images are continuous, high-dimensional, and spatially informative. These fundamental differences in representation and attributes can cause confusion and interference when directly merged.
- **Scale and Distribution Variance:** Text and image features often vary in numerical scale and distribution. Without appropriate scaling or standardization, this disparity can lead to uneven weighting in the model, affecting its decision-making capabilities.
- **Redundancy and Conflict:** Feature engineering may inadvertently introduce redundant or conflicting features. For example, overlapping information between text and image features might lead the model to overemphasize this shared information during training, potentially neglecting other vital features.

To confront these challenges, this paper focuses on developing an Adaptive Feature Matching Optimization framework (AFMO) for fake news detection with multimodal resources. AFMO integrates both textual and visual modalities to construct a semantically enriched and structurally robust representation of news content. A key innovation of AFMO lies in the use of a simulated annealing algorithm for feature fusion. Unlike prior works that treat multimodal features as flat concatenations or rely on rigid attention mechanisms, our method dynamically selects and combines informative features across modalities in an optimization-aware manner. This design not only mitigates interference from visually dominant but semantically irrelevant cues, but also enhances the discriminative quality of the fused representation. From a theoretical perspective, AFMO contributes to the growing discourse on integrating AI and optimization by embedding meta-heuristics into representation learning. From a practical standpoint, AFMO improves the robustness of fake news detection systems deployed in real-world settings, where noisy, deceptive, or misleading visual-textual pairings are prevalent. Furthermore, in contrast to conventional approaches that treat all input data equally or overlook modality-specific noise, AFMO distinguishes itself through a selective, optimization-driven feature handling process.

Specifically, the main contributions of this research are as follows:

- 1) A principled anomaly detection strategy is proposed to identify and remove inconsistencies between textual and visual representations in news content, thereby enhancing the reliability and robustness of multimodal learning.
- 2) A selective feature fusion mechanism based on simulated annealing is developed to mitigate cross-modal interference and optimize the integration of heterogeneous modalities, resulting in improved detection accuracy.
- 3) The proposed techniques are embedded within a unified framework incorporating pre-trained deep feature extractors, and their effectiveness is empirically validated on multiple real-world benchmark datasets.

The remainder of this paper is organized as follows: [Section 2](#) discusses related work on fake news detection, with a particular focus on multimodal architectures. [Section 3](#) presents the proposed AFMO in detail. [Section 4](#) describes the dataset used and presents an analysis of baselines, along with experimental results. Finally, [Section 5](#) concludes the paper.

2. Related work

In today's digital age, we are incessantly bombarded with an overwhelming amount of information, much of which is shaped by online platforms. This proliferation of digital content, while representing a vast source of knowledge, also presents a significant

challenge due to the rampant dissemination of fake news (Vosoughi et al., 2018; Weismueller et al., 2024). The widespread distribution of misinformation can have profound societal consequences, as evidenced by the public health risks posed during the COVID-19 epidemic (Alghamdi et al., 2023). Thus, the imperative to detect and counter these falsehoods has become a pivotal issue, marking a critical juncture in managing digital content. Traditionally, fake news detection has predominantly relied on unimodal techniques that primarily focus on single data sources, such as text or images. However, these methods often fall short by failing to account for the rich and multifaceted nature of digital content, which inherently combines various modalities (Singhal et al., 2019). Recognizing this limitation, recent research has begun exploring the potential of multimodal information, integrating multiple data types to enhance the effectiveness of fake news detection. However, the optimal strategies for combining these diverse modalities, selecting relevant features, and optimizing detection models remain relatively unexplored territories.

2.1. State-of-the-art unimodal fake news detection

The study of misinformation, particularly its detection and correction, is critical in the age of digital media. Recent literature highlights various strategies to address this challenge, focusing on concepts such as prebunking and debunking. Moravec et al. (2020) explore interventions based on cognitive theories that appeal to both intuitive and analytical thinking processes, demonstrating how these can be leveraged to enhance the detection of fake news on social media platforms. Similarly, Roozenbeek et al. (2022) discuss the effectiveness of “psychological inoculation”, which prepares individuals to resist misinformation by exposing them to weakened forms of the arguments before they encounter them in the wild. Ecker et al. (2022) further contribute to this field by detailing the psychological drivers behind the belief in misinformation of such beliefs even when corrections are presented. Together, these studies form a crucial foundation for understanding the complexities of misinformation and highlight the need for a nuanced approach to its detection and correction in unimodal contexts.

In the past decade, considerable efforts have been devoted to detecting fake news using machine learning methods (Farhangian et al., 2024; Qu et al., 2024). Specifically, two main categories are distinguished: those based on traditional machine learning methods (Ruchansky et al., 2017) and those based on deep learning methods (Ma et al., 2016). In previous studies, text and user information were predominantly used to extract text features through statistical machine learning or neural networks. For example, linguistic features were manually selected and solely text information was employed for classification (Ruchansky et al., 2017). However, early studies on fake news detection often relied on manually extracting relevant text information and handcrafting features, which were then fed into the aforementioned machine-learning models for classification. For instance, Castillo et al. (2011) manually extracted features such as punctuation marks and word count as indicators of news authenticity.

In light of recent strides in deep learning methodologies, neural network models have ascended as the prevailing approach for fake news detection (Ma et al., 2019). Researchers have harnessed architectures such as CNN (Yu et al., 2017) and RNN (Ma et al., 2016) to dissect and unearth falsified information. Notwithstanding their promising outcomes, the lion's share of investigations has predominantly concentrated on textual attributes, often disregarding the latent advantages of incorporating image features (Su et al., 2019). In addition, some scholars use a generative approach (Yang et al., 2019) to do fake news detection. However, within the realm of social media, news articles often traverse broader spheres when accompanied by visual content (Ford et al., 2022), owing to their visual allure and the multifaceted ideas they encapsulate.

2.2. State-of-the-art multimodal fake news detection methods

Nevertheless, the aforesaid methodologies predominantly apply single-modal data for fake news detection. Consequently, numerous researchers have shifted their focus towards the inclusion of images in counterfeit news detection, resulting in the proposition of multimodal detection frameworks (Qi et al., 2019). In the realm of multimodal fake news detection, systematic research experiments were conducted by Jin et al. (2017) to assess the role of images. Impressive results were achieved through an attention-based RNN multimodal fusion framework, effectively incorporating information from both textual and visual sources. A significant contribution, EANN, accentuates cross-domain fake news detection by eliminating domain-specific textual and visual information, while learning common feature representations across diverse domains (Wang et al., 2018). In a parallel vein, a 2019 study introduces a memory network module that engenders invariant features between different events for cross-domain recognition (Zhang et al., 2019). Diverging slightly, MVAE follows the blueprint of EANN, albeit its core architecture rests upon the foundation of Variational Autoencoder (VAE) utilizing word embedding vectors. It employs bidirectional LSTM to distill text and image representations from a pre-trained VGG-19 model. Subsequently, the concatenated hidden vector undergoes decoding to reconstruct the original samples. Further, these hidden vectors traverse two fully connected layers to facilitate fake news detection (Khattar et al., 2019).

Recent advances in multimodal fake news detection have applied deep learning techniques to integrate heterogeneous information from textual and visual modalities. Among the most representative approaches, BMR (Bootstrapping Multi-view Representations) (Ying et al., 2023) combines multi-view feature extraction with a multi-gate mixture-of-experts mechanism to enhance the alignment and consistency of cross-modal representations. Despite its effectiveness, BMR relies on a fixed fusion structure, which may be vulnerable to semantic interference, especially in noisy or ambiguous media environments. To improve alignment flexibility, MRML (multimodal Rumor Detection by Metric Learning) (Peng et al., 2023) adopts a deep metric learning paradigm that simultaneously captures intra-modal and inter-modal relationships through triplet-based and contrastive learning strategies.

From a semantic perspective, CSFND (Peng et al., 2024) integrates local contextual and global semantic representations to distinguish fake news based on hierarchical textual reasoning. To address representation bias from a frequency-domain angle, FSRU (Lao et al., 2024) applies Fourier transforms to convert spatial features into spectral signals, which are then processed via dual contrastive

learning to derive discriminative representations. However, the neglect of spatial contextual cues may hinder its performance in tasks requiring fine-grained visual grounding. In a more adaptive design, Miner-uvs (Wang et al., 2024) introduces a modality perception learning strategy that estimates unimodal veracity signals to guide feature weighting during fusion. This approach effectively mitigates modality conflicts but relies heavily on the accuracy of individual modality assessments.

A growing body of work on multimodal fake news detection moves beyond shallow concatenation or generic attention by explicitly modeling inconsistency between textual and visual evidence. MFIR (Wu et al., 2023b) proposes a fusion-and-reasoning framework that learns inconsistent semantics to support explainable decisions, coupling cross-modal fusion with inconsistency learning and an evidence-oriented reasoning layer. MRHFR (Wu et al., 2023a) adopts a cognition-inspired perspective and introduces multi-reading-habits fusion reasoning, using distinct reading patterns as inductive biases for deeper integration and coherence-aware inference. Similarly, HCCIN (Wu et al., 2024) formulates detection as a human cognition-motivated consistency inference process, combining cross-modal alignment with consistency reasoning to exploit both consistent and inconsistent cues.

The emergence of large language models (LLMs), such as GPT-4 (Achiam et al., 2023) and GPT-4o (OpenAI et al., 2024), has substantially advanced natural language understanding. Their strong multilingual and multimodal capabilities make them attractive for information verification. However, recent work has also exposed clear limitations in domain-specific and multimodal fake news detection, where adaptation to evolving misinformation patterns remains difficult without task-specific optimization. To address this issue, Hu et al. (2025) proposed GLPN-LLM, which incorporates LLM knowledge through global label propagation rather than treating LLM outputs as direct predictions. Their results indicate that structured integration of LLM-derived semantic priors with task-specific relational modeling is more effective than relying on LLM predictions alone. Along this line, Linfeng et al. (2026) proposed FVDLM, a fake news video detection framework that augments LLMs with cross-modal retrieval of external evidence. Instead of relying only on the target content, FVDLM retrieves related videos and textual news to enlarge the evidential basis for verification. It further employs an information bottleneck to suppress irrelevant video noise and a prompt-based LLM module to assess consistency between the retrieved evidence and the target item. From a complementary perspective, Zhu et al. (2025) advanced multimodal fake news detection at both the dataset and modeling levels. They introduced MFND, a benchmark for highly realistic manipulated news with 11 manipulation types and support for both detection and localization. They proposed a Shallow-Deep Multitask Learning (SDML) framework that jointly exploits unimodal and cross-modal information. At the shallow stage, SDML performs fine-grained image-text alignment via momentum-distillation-based contrastive learning and adaptive cross-modal fusion; at the deep stage, it adopts a two-branch architecture to enhance image and text representations before integrating them with mutual-modal features for final prediction.

Distinct from prior work, this study introduces an approach that harnesses the interpretability of operations research techniques to systematically mitigate cross-modal interference. By integrating these techniques into the feature selection and fusion process, the proposed method enhances both the computational efficiency and detection reliability in multimodal fake news identification.

2.3. Training enhancement

There are various approaches to improving the quality of training, which can be broadly categorized into two perspectives: the data perspective and the usage perspective. From the data perspective, improvements can be achieved through data preprocessing, outlier detection, and data augmentation. In this paper, we focus on the outlier detection-based approach. Outlier detection is a method used to identify anomalous behavior that deviates from the expected patterns within the original dataset. It can be classified based on the techniques employed, including statistical-based methods, distance-based methods, clustering-based methods, and deep learning-based methods. Statistical methods typically utilize metrics such as mean, standard deviation, median, and quartiles to determine outliers that fall outside predefined thresholds (Hodge & Austin, 2004). Distance-based methods (Orair et al., 2010) and density-based methods (Breunig et al., 2000) effectively detect isolated outliers by measuring the distances or densities between data points. Clustering methods (Duan et al., 2009) leverage the inherent clustering structure of data points to detect outliers that do not belong to any specific cluster or are significantly distant from other clusters. Deep learning-based methods for anomaly detection often employ generative adversarial networks, where the models are trained through an adversarial process between generators and discriminators to learn the data distribution and features. The generators are then utilized to generate new samples, which are used for anomaly detection (Akcaay et al., 2019).

Mahalanobis distance and k-nearest neighbors (KNN) are two widely used and effective outlier detection algorithms due to their simplicity and efficiency. Mahalanobis distance has found frequent application in detecting abnormal Distributed Denial of Service (DDoS) attacks and related fields (Daneshgadeh Çakmakçı et al., 2020). On the other hand, KNN is commonly employed in regression and classification problems, such as modeling optimal temperature adjustments for users using KNN (Xiong & Yao, 2021). Considering the demonstrated effectiveness of Mahalanobis distance and KNN outlier detection in various applications, we incorporate these two methods into the proposed AFMO framework to enhance the reliability of the training process.

2.4. Simulated annealing

The simulated annealing algorithm, which is initially introduced by Kirkpatrick (1984), stands as a versatile stochastic search technique extensively employed across a spectrum of combinatorial optimization problems (Suman & Kumar, 2006). Simulated annealing has been effectively integrated with other computational methods to achieve superior performance compared to standalone approaches, particularly in domain-specific optimization tasks. For instance, the convergence of simulated annealing with genetic

Table 1
Description of mathematical notations.

Variables	Explanation
$D = \{(txt_i, img_i, label_i), i = 1, \dots, N\}$	News dataset with N elements.
txt_i	The text of the i -th news item.
img_i	The image of the i -th news item.
$label_i$	The label of the i -th news.
$m^t = \{m_i^t \mid \forall i = 1, \dots, N\}$	Text features of all news.
$m^p = \{m_i^p \mid \forall i = 1, \dots, N\}$	Image features of all news.
$D_{reduce} \subset D$	Training set after removing outliers.
$m_i^t = \{f_j^t \mid \forall j = 1, \dots, n_{txt}\}$	Value of the j -th dimension in the text feature.
$m_i^p = \{f_j^p \mid \forall j = 1, \dots, n_{img}\}$	Value of the j -th dimension in the image feature.
$x = (x_1, \dots, x_j, \dots, x_n)$	Indicates which text features were selected.
$y = (y_1, \dots, y_j, \dots, y_n)$	Indicates which image features were selected.
$m_i^t \subseteq m_i^t \oplus m_i^p$	Selected features after processing.

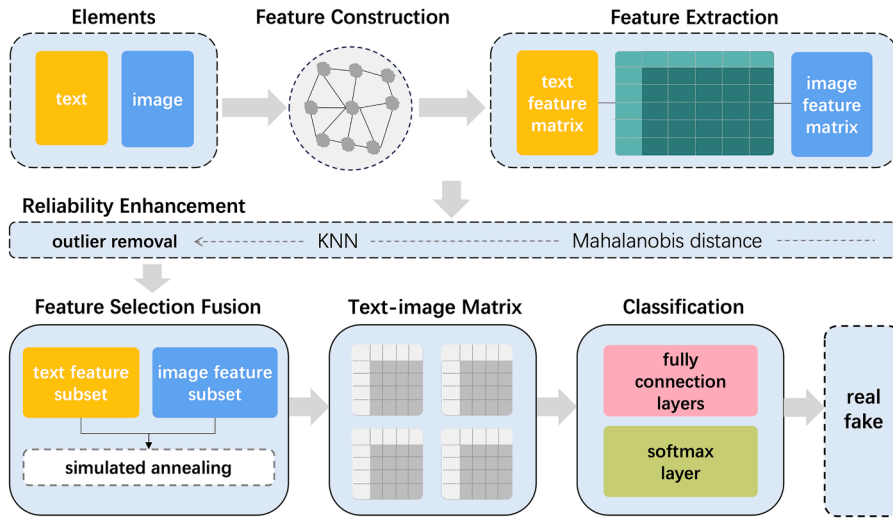


Fig. 2. Refined multimodal fake news detection using BERT and VGG for modality-specific feature extraction, outlier-aware filtering, and adaptive feature fusion using simulated annealing for improved cross-modal representation and classification accuracy.

algorithms has been harnessed to solve the multi-class multidimensional knapsack optimization problem (Meng et al., 2019). Furthermore, the amalgamation of simulated annealing with the Salp Swarm Algorithm (SSA) and genetic algorithm has been wielded to calibrate the equilibrium between exploration and exploitation within SSA (Kassaymeh et al., 2022). Motivated by the proven effectiveness of simulated annealing in combinatorial optimization, we incorporate it into the proposed AFMO framework to address the complex challenge of synergistically integrating textual and visual features in a multimodal setting.

3. Methodology

3.1. Overview of AFMO

The principal formulas employed in this article are delineated in Table 1.

Consider a news dataset D comprising N samples, each consisting of a news text, news image, and corresponding true or false label. The proposed AFMO primarily applies the textual and image information of the news to infer its label.

As shown in Fig. 2, AFMO first extracts modality-specific representations from each news item using BERT for text and VGG for images. To improve data reliability, we then detect and remove anomalous training instances using Mahalanobis distance and KNN, yielding a reduced dataset D_{reduce} . On this curated set, simulated annealing performs adaptive feature selection by jointly choosing informative dimensions from the text features m^t and image features m^p and fusing them into a compact multimodal representation. The fused vector is finally passed through a lightweight stack of fully connected layers to produce the prediction.

Unlike fusion schemes based on direct concatenation or attention, AFMO treats multimodal fusion as a feature-selection problem and solves it via a principled stochastic combinatorial search. This explicit selection suppresses redundant or interference-prone dimensions, thereby mitigating cross-modal interference and improving robustness under heterogeneous conditions. Conceptually, AFMO (i) casts cross-modal interference mitigation as an optimization problem over feature subsets, and (ii) increases interpretability and controllability by making fusion decisions explicit through feature masks rather than implicit end-to-end fusion weights.

3.2. Feature extraction

3.2.1. Text feature extraction

Text feature extraction techniques include TF-IDF (Term Frequency-Inverse Document Frequency), Word2Vec (a word vector representation method encompassing skip-gram and Continuous Bag of Words (CBOW) training [Mikolov et al., 2013](#)), and GloVe (Global Vectors for Word Representation, a word vector representation method trained using word co-occurrence matrix in the corpus [Pennington et al., 2014](#)). However, these conventional text feature representation methods are unable to capture word vectors that vary based on context and fail to address the issue of word polysemy. For instance, the word “bank” can refer to either a financial institution or a river bank in different contexts. None of the aforementioned methods can distinguish between these meanings, resulting in identical text features despite their distinct semantics. To address this limitation, BERT (Bidirectional Encoder Representations from Transformers) ([Devlin et al., 2019](#)) is employed as a text feature extraction method. BERT, through extensive pre-training on large corpora and a sophisticated attention mechanism, can discern the diverse semantic interpretations of the same words in different contexts, thereby providing a more nuanced representation of the text features.

BERT (Bidirectional Encoder Representations from Transformers) ([Devlin et al., 2019](#)) stands as a transformative model grounded in the principles of attention mechanisms, and it is pre-trained via the Masked Language Model (MLM) task. In our investigation, we designate txt_i as the input for BERT, given that the terminal layers of BERT encapsulate a comprehensive textual mosaic. Empirical findings have consistently demonstrated that the outcomes of the last quartet of hidden layers offer optimal performance across diverse NLP tasks, in contrast to the results generated by other layers. Accordingly, we concatenate the outputs from these final four hidden layers, thereby cultivating elemental textual features, resulting in a cumulative tally of four outputs. These outputs are sequentially concatenated to form the complete basic text feature, denoted as m_i^t , as shown in the following equation:

$$m_i^t = TokenEmbeddings(txt_i) + PositionEmbeddings(txt_i) + SegmentEmbeddings(txt_i).$$

Here, m_i^t represents the embedding vector of the i -th text sample, $TokenEmbeddings$ corresponds to the initial embedding vector of the token, $PositionEmbeddings$ represents the position embedding vector of the token, and $SegmentEmbeddings$ indicates the segment embedding vector of the token.

3.2.2. Image feature extraction

Image features are extracted using VGG (Visual Geometry Group) ([Simonyan & Zisserman, 2014](#)). With its deep network architecture, VGG exhibits enhanced learning capability in capturing image features. The utilization of numerous small-sized convolutional kernels in the network enables the VGG model to converge quickly with a minimal number of iterations, thereby improving training efficiency.

Within our study, we employ VGG-19, pre-trained on the ImageNet dataset, as the designated image information feature extractor. When presented with an input image of dimensions $W \times H$, a sequence of convolutional and pooling layers collaborates to engender a feature vector of dimension C , encapsulating the salient image characteristics. In this context, C corresponds to the channel count within the ultimate convolutional layer. Within our methodology, img_i assumes the role of input for VGG-19. Aligned with the structural blueprint of the VGG network, the terminal layer's output encapsulates an all-encompassing spectrum of image traits. As a result, we adopt the terminal layer's output from VGG-19 as our image features, a representation realized through the following equation:

$$m_i^p = VGG19(img_i).$$

3.3. Reliability enhancement

The text features m_i^t and image features m_i^p of each sample are concatenated to form m_i , where $m_i = m_i^t \oplus m_i^p$. Consequently, all the features of the samples are represented as $M = \{m_i \mid \forall i = 1, \dots, N\}$.

3.3.1. Mahalanobis distance

The Mahalanobis distance serves as a conventional metric to gauge the separation between a specific point and a distribution. In our methodology, we apply the Mahalanobis distance computation to discern and subsequently eliminate anomalous data instances. Indeed, we embark on deriving the mean vector μ and the covariance matrix Σ of the feature set M . Thereafter, the distance between each sample and the dataset as an aggregate entity is computed. The threshold, denoted as $Dist_{mean}$, is ascertained by calculating the average distance across all samples and then multiplying it by the standard deviation σ , adjusted by a weighting factor α . The default value for the weighting coefficient α is set at 1. Instances, where the distance between a sample point and the dataset surpasses this threshold, are deemed abnormal and are subsequently excluded from consideration.

The Mahalanobis distance $Dist(m_i)$ is computed as follows:

$$Dist(m_i) = \sqrt{(m_i - \mu)^T \Sigma^{-1} (m_i - \mu)}.$$

The threshold for abnormality, $Dist_{max}$, is obtained by adding the product of α and σ to $Dist_{mean}$:

$$Dist_{max} = Dist_{mean} + \alpha \cdot \sigma.$$

3.3.2. K-nearest neighbor

KNN (k-nearest neighbors) emerges as a prevalent unsupervised clustering algorithm, seamlessly extending its utility to encompass outlier detection. The foundational tenet revolves around gauging the distance between the present sample and the entirety of the dataset. Divergent from quantifying distances between points and distributions, KNN computes distances between each individual sample and the remaining samples, employing the Euclidean distance formula, illustrated through the ensuing equation:

$$\text{dist}(m_i, m_j) = \left(\sum_{l=1}^n (m_i^l - m_j^l)^2 \right)^{\frac{1}{2}}.$$

The k samples characterized by the closest distances are meticulously chosen, and the arithmetic mean of the distances between the current sample and these k samples is meticulously computed. These average distances are then amassed and ordered in descending order, ranging from the largest to the smallest values. By delineating a threshold to demarcate the proportion of outliers, we proceed to ascertain whether the average distance associated with the current sample surpasses this predefined threshold. Should the average distance exceed the set threshold, the current sample is categorically categorized as an outlier. During the ultimate phase, these outlier samples are systematically excised, culminating in the inception of a fresh dataset termed D_{reduce} . This distilled dataset subsequently forms the foundation for subsequent training and testing phases, employing the identical test set utilized within the original dataset.

3.3.3. MD-KNN outlier detection algorithm

Algorithm 1 describes our conservative outlier-filtering procedure, designed to improve training data reliability while minimizing the risk of discarding informative hard samples. Initially, the dataset D is divided randomly into five equal subsets D_k , where $k = 1, \dots, 5$. This division facilitates cross-validation, allowing each subset to serve as a test set while the remaining subsets are used for training. For each subset D_k , we detect anomalous instances in the training pool using two criteria: MD, which measures how far a sample deviates from the overall feature distribution, and KNN, which captures local neighborhood irregularities. Rather than removing all samples flagged by either detector, the intersection of outliers detected by both methods is then removed from D_k , resulting in a reduced subset D'_k . This intersection strategy is intentionally precision-oriented: only samples that are consistently deemed suspicious by both a global MD and a local KNN detector are excluded, thereby improving robustness and reducing the chance of over-filtering.

After processing all subsets, each D_k serves once as a test set while the remaining four reduced subsets D'_k are combined to form a training set. The algorithm then calculates the accuracy of the model on each test set D_k , denoted by Acc_k . The mean accuracy across all subsets is computed to evaluate the overall performance of the outlier detection settings. Using the best parameter settings determined earlier, the algorithm applies the outlier detection methods again to each sub-dataset D_k . This step ensures that the refined parameters are used to achieve the most accurate and reliable training dataset possible.

Algorithm 1 MD-KNN outlier detection.

Require: Dataset D ;

Ensure: A reduced training sets;

- 1: Split D randomly into five equal subsets: $(D_k), k = 1, \dots, 5$;
 - 2: **for** each D_k **do**
 - 3: Apply both MD and KNN to identify outliers;
 - 4: Let D'_k be a reduce set by removing the intersection of outliers detected by MD and KNN from D_k ;
 - 5: **end for**
 - 6: **for** each test set D_k **do**
 - 7: Combine the remaining four sub-datasets (i.e., D'_k) to form a training set;
 - 8: Calculate the accuracy on its corresponding test set D_k , noted by Acc_k ;
 - 9: **end for**
 - 10: Calculate the mean accuracy according to the $\text{Acc}_k, k = 1, \dots, 5$;
 - 11: Determine the best parameter setting for both MD and KNN, maximizing the average accuracy;
 - 12: **for** each sub-dataset D_k **do**
 - 13: Apply outlier detection algorithms with the best parameter setting to identify outliers in D_k ;
 - 14: **end for**
-

3.4. Feature selection fusion

As presented in Fig. 2, the text and image information of the news are extracted using the feature extraction module consisting of BERT and VGG-19, respectively, to obtain the text and image representations of the news. Subsequently, the outlier removal module eliminates samples with abnormal features, resulting in a new training set. Finally, the information from different dimensions of the text and image modalities is selected and fused using the simulated annealing algorithm. These fused features are then fed into the classifier to obtain the final results. Within the original dataset, each text feature m_i^t is epitomized by a collection of values $\{f_1^t, f_2^t, \dots, f_{n_{txl}}^t\}$, wherein n_{txl} denotes the extent of the text feature. In parallel, every image feature m_i^p is characterized by a set of

values $\{f_1^p, f_2^p, \dots, f_{n_{img}}^p\}$, with n_{img} reflecting the dimensionality of the image feature. Traditional feature selection tactics encompass the aggregation of the two modality features, a practice that may inadvertently culminate in the loss of information, particularly when the feature vectors embody divergent connotations. Nonetheless, this approach exponentially escalates computational complexity and may inadvertently incorporate irrelevant or spurious information from all dimensions, thereby potentially compromising outcomes.

To address this problem, we model the feature selection problem as a binary feature-subset optimization (a combinatorial selection problem) and solve it with simulated annealing, poised to sieve out efficacious dimensions whilst discarding those devoid of relevance. Let $x \in \{0, 1\}^{n_{txt}}$ and $y \in \{0, 1\}^{n_{img}}$ be binary variables indicating whether each text/image dimension is retained. We seek solutions that maximize the downstream classification performance, so dimensions that introduce redundancy or conflicting cues are suppressed because they decrease the objective:

$$\max_{x,y} \text{Acc}(\text{detector}((x \cdot m^t, y \cdot m^p))) \quad \text{s.t.} \quad x \in \{0, 1\}^{n_{txt}}, y \in \{0, 1\}^{n_{img}}. \quad (1)$$

The simulated annealing algorithm exhibits the capacity to transcend local optima, thereby accommodating suboptimal solutions during the exploration process and evading the entrapment within local optima. Through this approach, we aspire to elevate the caliber of selected features, curtail possible interferences between textual and visual insights, and augment the overall efficacy of classification tasks. Formally, the main steps of the simulated annealing algorithm for feature selection are displayed in [Algorithm 2](#).

Algorithm 2 Simulated annealing-based feature selection.

Require: Feature set M .

Ensure: Selected feature set M_{selected} .

- 1: Set minimum temperature as $t_{\min}$, cooling factor as k , maximum iterations as max_iter ;
 - 2: Set the current temperature t_{cur} as t_0 ;
 - 3: Randomly generate an initial solution $x \oplus y$;
 - 4: **for** $i \leftarrow 1$ to max_iter **do**
 - 5: If $t_{\text{cur}} < t_{\min}$, terminate the iteration;
 - 6: Generate a new solution $x' \oplus y'$ based on $x \oplus y$;
 - 7: Apply $x' \oplus y'$ to M for generating M' ;
 - 8: Evaluate the performance of the new feature set M' ;
 - 9: Calculate the difference (ΔE) between the current and previous optimal accuracy rates;
 - 10: Calculate the acceptance probability $P(x' \oplus y' \rightarrow x \oplus y)$ based on t_{cur} and ΔE ;
 - 11: Determine whether to accept the new solution $x' \oplus y'$;
 - 12: If the new solution is accepted, update $x \oplus y$ to $x' \oplus y'$ and M to M' ;
 - 13: Decrease the current temperature: $t_{\text{cur}} = t_{\text{cur}} \cdot k$;
 - 14: **end for**
 - 15: **return** The selected feature set M_{selected} .
-

More preciously, [Algorithm 2](#) includes the following three modules:

- **Heating Up Process** (cf., Steps 1–3): Initially, an initial temperature t_0 , a minimal temperature $t_{\min}$, and a prevailing temperature t_{cur} (initially aligning with t_0) are predetermined. For every combination of $m^t \oplus m^p$, a binary sequence $x \oplus y$ is conceived, the length of which corresponds to the collective dimensionality of the encompassed features. Within this sequence, $x_i = 1$ (resp. $y_i = 1$) if the i -th dimension of the text (resp. image) feature is adjudged pertinent for the ultimate fake news detection, retaining its inherent value. In contrast, $x_i = 0$ (resp. $y_i = 0$) if the i -th dimension lacks significance. A completely arbitrary $x \oplus y$ configuration is initialized, serving as the foundational state.
- **Isothermal Process** (cf., Steps 5–8): If the current temperature t_{cur} is less than the minimum temperature $t_{\min}$, the iteration process draws to a close. Conversely, if this condition remains unmet, a fresh $x \oplus y$ configuration is engendered grounded in its predecessor. This metamorphosis is effectuated by electing a certain number of dimensions from the initial $x \oplus y$ arrangement and subsequently inverting their prevailing 0–1 values. The extent of randomness and the tally of selected dimensions escalates in tandem with elevated temperatures. This escalating relationship is mathematically formulated by the ensuing equation:

$$\text{length}_{\text{change}} = \frac{t_{\text{cur}} - t_{\min}}{t_0 - t_{\min}} \times (N_{\text{txt}} + N_{\text{img}}). \quad (2)$$

The updated $x \oplus y$ configuration is employed to transform the initial features m_i , yielding the reconfigured features m_i^r . Subsequently, these reconfigured features are introduced into the classifier detector, leading to the derivation of the classification outcome out_i . The accuracy rate is meticulously computed through the juxtaposition of out_i against the actual label label_i . This accuracy rate emerges as the pivotal objective function governing the simulated annealing algorithm:

$$\text{out}_i = \text{detector}(m_i^r), \quad (3)$$

where,

$$m_i^r = (x \cdot m^t, y \cdot m^p). \quad (4)$$

Table 2
The statistics of considered datasets.

Dataset	Fake News	Real News	Texts	Images
PolitiFact	463	373	836	-
Chinese Weibo	4108	3615	7723	7723
English GossipCop	3398	12365	15763	15,763

- **Cooling Process** (cf., Steps 9–13): If the accuracy rate of the current iteration is higher than the previous optimal accuracy, the revised $x \oplus y$ configuration is directly embraced. In contrast, if the accuracy rate falls below, the determination to embrace the fresh solution transpires probabilistically, hinging on the prevailing temperature t_{cur} and the discrepancy in accuracy between the existing and preceding optimal accuracy, represented by ΔE . Elevated temperatures engender heightened probabilities of adoption, effectively fostering the avoidance of local optima in favor of global optima. The probability of integrating the prevailing solution is governed by the ensuing equation:

$$P(x' \oplus y' \rightarrow x \oplus y) = e^{-k \frac{\Delta E}{t_{cur}}}. \quad (5)$$

If the current solution is accepted, its accuracy rate assumes the mantle of the optimal accuracy rate. Subsequently, the prevailing temperature is multiplied by the cooling factor denoted as k , and this updated temperature is funneled back into the Isothermal Process phase. In the event that the current solution is not assimilated, the progression reverts promptly to the Isothermal Process stage.

The above three steps are repeated multiple times to complete the multi-round simulated annealing algorithm, which aims to achieve the best results.

4. Experimental results

4.1. Description of dataset

In the context of unimodal testing, we opted to assess the efficacy of the proposed Adaptive Feature Matching Optimization (AFMO) framework using the PolitiFact dataset (Shu et al., 2020). Acquired from FakeNewsNet, the PolitiFact dataset encompasses news articles drawn from the fact-checking website PolitiFact, meticulously labeled for their authenticity by domain experts. For the multimodal scenario, we selected two extensively recognized and publicly accessible benchmark datasets—namely, the Chinese Weibo dataset and the English GossipCop dataset—for our rigorous evaluation and experimentation in the domain of fake news detection. To offer a concise overview of the dataset characteristics, we present a summary of the dataset statistics in Table 2. To assess the practical effectiveness and robustness of the proposed method, all input samples were used in their raw form without any preprocessing prior to training or inference.

4.2. Experiment setup

The division of the dataset into training, validation, and test sets follows a ratio of 7:1:2, ensuring an appropriate distribution. The computational resources harnessed for our experimentation encompass an Intel(R) Xeon(R) Gold 6248R CPU with a clock speed of 3.00GHz, coupled with an NVIDIA A100 80GB PCIe GPU.

Concerning the text modality, we apply the sentence-level vector outputs from the last four layers of the BERT model to serve as the textual feature representation. These representations undergo processing through a tandem of fully connected layers, culminating in the derivation of text features with a dimensionality of 64. Concurrently, for the image modality, the terminal layer's output within the FEATURES layer of the VGG-19 model assumes the role of image feature representation. Analogous to the text features, a solitary fully connected layer facilitates the creation of image features with a dimensionality of 64. To counteract potential overfitting, both the BERT and VGG-19 models incorporate parameter freezing. In addition, a dropout layer with a dropout rate of 0.5 is strategically interposed following each fully connected layer. The chosen batch size is 90, while the optimization scheme entails Adam with an initial learning rate established at 0.001. The training trajectory encompasses 100 epochs.

For the integrated simulated annealing algorithm, the initial temperature is set to $\exp(4)$, the minimum temperature is defined as $\exp(-1)$, and the temperature decay coefficient is stipulated at 0.98. Furthermore, the versions of the pivotal Python packages engaged in this research is provided as follows: python = 3.7.4, pytorch = 1.11.0, cuda = 10.2, torchvision = 0.11.2 + cu102.

4.3. Baseline models

We compare our approach with the state-of-the-art methods and some baselines, as listed below:

- **SAFE** (Zhou et al., 2020) explores inconsistencies between textual and visual content by identifying mismatches within news articles, offering an early approach to cross-modal inconsistency detection.
- **SpotFake+** (Singhal et al., 2020) employs transfer learning to capture deep semantic and contextual signals from both text and associated images, enabling the detection of fake news in full-length articles through a multimodal lens.

- **CAFE** (Chen et al., 2022) introduces cross-modal alignment combined with ambiguity modeling, allowing the model to adaptively integrate multimodal features based on varying levels of cross-modal ambiguity.
- **BMR** (Ying et al., 2023) enhances cross-modal consistency prediction by extracting representations from textual content, image patterns, and visual semantics, and refining them using a Multi-gate Mixture-of-Experts network.
- **MRML** (Peng et al., 2023) incorporates deep metric learning to jointly model intra-modal and inter-modal relationships, utilizing triplet loss within modalities and contrastive pairwise loss across modalities to improve rumor detection.
- **CSFND** (Peng et al., 2024) proposes a contextual semantic fusion strategy that combines local context with global semantics, enabling the model to better differentiate fake from real news through semantic dissimilarity.
- **FSRU** (Lao et al., 2024) applies a Fourier-based spectral transformation to represent spatial features in the frequency domain, followed by dual contrastive learning to extract discriminative spectrum-aware multimodal representations.
- **Miner-uvs** (Wang et al., 2024) mitigates cross-modal conflicts by introducing unimodal veracity signals, which estimate the truthfulness of individual modalities and use these signals to reweight features during fusion, reducing the neutralization effect.
- **GPT-4o** (OpenAI et al., 2024), developed by OpenAI, represents a major advancement in large language models, demonstrating strong reasoning and contextual generation capabilities across diverse tasks, and providing a foundation for emerging multimodal applications.
- **GLPN-LLM (HMCAN) and GLPN-LLM (CLIP)** (Hu et al., 2025) incorporate LLM-derived supervision through label propagation, while adopting HMCAN (Qian et al., 2021) and CLIP (Radford et al., 2021) as their underlying multimodal encoders, respectively.

In this study, we employ Accuracy, Precision, Recall, and F1 score as evaluation metrics to assess the model's performance. Accuracy represents the percentage of correct predictions made by the model. Precision measures the percentage of correct predictions among the samples that are predicted as positive by the model. Recall calculates the percentage of all positive samples that the model correctly identifies. F1 score is a weighted average of Precision and Recall.

4.4. Result and analysis

4.4.1. Comparative analysis

To evaluate the effectiveness of the proposed **AFMO** framework, we conducted a comprehensive comparative analysis against a range of representative multimodal fake news detection approaches on three widely adopted benchmark datasets: **Weibo**, **GossipCop**, and **PolitiFact**. The results, in terms of classification accuracy, are reported in Table 3. Given that the proposed AFMO framework employs "Accuracy" as the explicit optimization objective within Algorithm 2, we adopt accuracy as the sole evaluation metric for comparative analysis. This choice ensures methodological consistency between the model's internal objective function and external performance assessment. While other metrics such as precision, recall, and F1-score will be employed in the ablation experiments analysis to evaluate the robustness of AFMO.

Based on the results presented in Table 3, several key observations can be made regarding the comparative performance of the evaluated methods.

- AFMO achieves the highest accuracy on both **Weibo (0.923)** and **PolitiFact (0.917)**, outperforming all existing methods, including the strongest baselines BMR and Miner-uvs (each at 0.914).
- Conventional models such as SAFE and SpotFake+ achieve relatively lower accuracy, primarily due to limited modelling of cross-modal dependencies. More advanced models (e.g., BMR, MRML, FSRU) have incorporated multi-view learning, deep metric learning, and frequency-domain analysis to enhance multimodal representation. Nonetheless, these models still struggle with preserving cross-modal coherence and often suffer from semantic interference. In contrast, AFMO introduces an adaptive optimization strategy based on simulated annealing, which dynamically selects salient fused features while suppressing noisy or conflicting modality signals.
- AFMO also outperforms the recent GLPN-LLM variants on the Weibo dataset. Although GLPN-LLM(HMCAN) and GLPN-LLM(CLIP) both achieve an F1 score of 0.915, their Accuracy values are only 0.907 and 0.867, respectively, indicating weaker overall classification balance than AFMO. Notably, GLPN-LLM(CLIP) attains the highest Recall (0.933), exceeding AFMO (0.901), which suggests a more aggressive tendency to identify positive samples. However, this higher Recall is accompanied by substantially lower Accuracy and Precision, implying a higher false-positive rate. By contrast, AFMO does not maximize Recall, but achieves a more favourable balance among Accuracy, Precision, and F1. In fake news detection, such balanced performance is generally more desirable, since excessive false alarms can undermine both the credibility and the practical utility of the system.

The experimental findings show that the integration of deep neural feature learning with OR-inspired optimization techniques enables a more accurate and explainable solution to multimodal fake news detection.

4.4.2. Ablation experiments analysis

To assess the contribution of each component in AFMO, we perform an ablation study on the Weibo dataset (Table 4). We consider a baseline that concatenates BERT and VGG-19 features, and then add (i) outlier filtering, (ii) simulated annealing (SA) feature selection, or (iii) a random-search control that uses a constant temperature (i.e., random feature perturbations without annealing).

The baseline achieves 90.8% accuracy. Outlier filtering alone improves accuracy to 91.3%, indicating that removing anomalous training instances increases data reliability. Replacing the fusion step with SA further boosts accuracy to 91.9%, whereas the random-search control degrades performance (89.9%), suggesting that the gains are not due to arbitrary stochastic exploration but to the

Table 3
Comparison between AFMO and different methods on Weibo, GossipCop and PolitiFact datasets.

Dataset	Model	Year	Accuracy	Precision	Recall	F1
Weibo	SAFE	2020	0.816	0.817	0.816	0.817
	Spotfake +	2020	0.870	0.871	0.870	0.870
	CAFE	2022	0.840	0.840	0.840	0.840
	BMR	2023	0.914	0.914	0.914	0.913
	MRML	2023	0.897	0.897	0.896	0.896
	CSFND	2024	0.895	0.896	0.896	0.895
	FSRU	2024	0.901	0.901	0.903	0.901
	Miner-uvs	2024	0.914	0.914	0.914	0.914
	GPT-4o	2024	0.809	0.822	0.813	0.818
	GLPN-LLM(HMCAN)	2025	0.907	0.891	0.922	0.915
	GLPN-LLM(CLIP)	2025	0.867	0.898	0.933	0.915
	AFMO	–	0.923	0.947	0.901	0.923
GossipCop	SAFE	2020	0.838	0.836	0.854	0.839
	Spotfake +	2020	0.858	–	–	–
	CAFE	2022	0.867	0.853	0.864	0.848
	BMR	2023	0.881	0.878	0.880	0.871
	MRML	2023	0.831	0.817	0.831	0.809
	CSFND	2024	0.847	0.845	0.849	0.822
	FSRU	2024	0.849	0.840	0.852	0.838
	Miner-uvs	2024	0.884	0.879	0.884	0.877
	AFMO	–	0.875	0.885	0.971	0.926
PolitiFact	SAFE	2020	0.874	–	–	–
	Spotfake +	2020	0.856	–	–	–
	CAFE	2022	0.838	0.808	0.748	0.769
	BMR	2023	0.905	0.906	0.906	0.906
	MRML	2023	0.878	0.879	0.878	0.879
	CSFND	2024	0.851	0.854	0.852	0.851
	FSRU	2024	0.839	0.838	0.835	0.832
	Miner-uvs	2024	0.905	0.906	0.906	0.906
	AFMO	–	0.917	0.913	0.887	0.900

Table 4
Performance after adding different modules into AFMO.

Method	Accuracy	Precision	Recall	F1
Base(BERT + VGG-19)	0.908	0.909	0.913	0.911
Base + outlier	0.913	0.921	0.910	0.916
Base + random	0.899	0.900	0.901	0.901
Base + SA	0.919	0.925	0.918	0.922
Base + outlier + random	0.903	0.908	0.910	0.906
Base + outlier + SA	0.923	0.947	0.901	0.923

structured annealing schedule. Combining outlier filtering with SA yields the best results (92.3% accuracy), showing that the two modules are complementary: filtering reduces data noise, while SA selects a compact and discriminative feature subset for fusion.

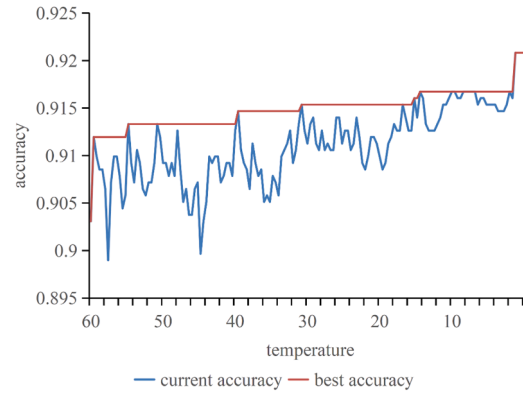
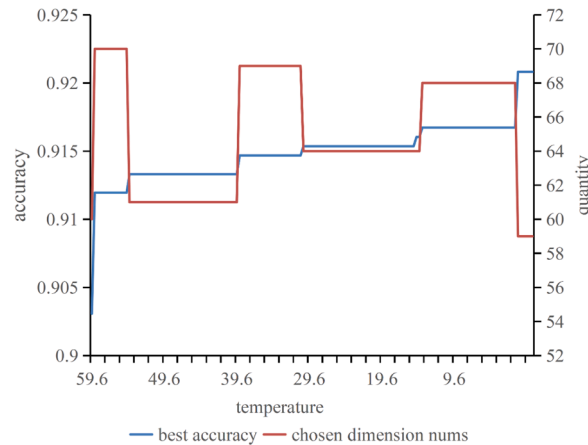
Table 5 shows that BERT layer aggregation strongly affects both SA-based feature matching and multimodal fusion in AFMO. Here, RateFS denotes the feature sparsity rate, and Runtime denotes the total optimization time accumulated over 100 runs with completely independent random seeds. In our framework, BERT first produces textual representations, which are then fused with visual features and refined by simulated annealing to select a compact, discriminative multimodal subset. The two aggregation schemes serve different purposes. “Concat” stacks hidden-layer vectors and thus preserves layer-specific information and inter-layer complementarity, but at the cost of rapidly increasing dimensionality. “Mean” averages the selected layers into a fixed-dimensional vector, yielding a more compact and stable representation while suppressing part of the heterogeneity across layers. The choice between them therefore directly influences both predictive performance and computational cost.

From Table 5, we observe that AFMO-BERT(Concat1) yields the weakest result (Accuracy = 0.902, F1 = 0.902), indicating that the final BERT layer alone does not provide sufficient semantic diversity for reliable downstream matching. Adding upper layers improves performance: AFMO-BERT(Concat2) raises both Accuracy and F1 to 0.913, suggesting that adjacent high-level layers provide complementary task-relevant information. The best result is obtained by AFMO-BERT(Concat4) (Accuracy = 0.923, Precision = 0.947, Recall = 0.901, F1 = 0.923), which implies that concatenating the last four layers offers the best balance between representational richness and manageable redundancy. Under this setting, SA can exploit inter-layer complementarity without facing an excessively large search space.

To assess computational cost, we measure only the runtime of the SA-based feature-matching stage, i.e., the phase in which AFMO repeatedly generates feature masks and performs forward propagation in CNN Fusion. By excluding data loading and the feature

Table 5The ablation of concatenating the last i hidden layers of BERT, $i = 1, 2, 4, 6, 8$.

Method	Accuracy	Precision	Recall	F1	RateFS	Runtime
AFMO-BERT(Concat1)	0.902	0.903	0.903	0.902	97.96%	70.83
AFMO-BERT(Concat2)	0.913	0.914	0.913	0.913	89.24%	90.61
AFMO-BERT(Mean2)	0.911	0.911	0.911	0.911	89.25%	71.82
AFMO-BERT(Concat4)	0.923	0.947	0.901	0.923	97.99%	127.67
AFMO-BERT(Mean4)	0.909	0.910	0.909	0.909	97.96%	71.93
AFMO-BERT(Concat6)	0.914	0.914	0.914	0.914	49.30%	165.29
AFMO-BERT(Mean6)	0.913	0.913	0.912	0.913	88.69%	71.96
AFMO-BERT(Concat8)	0.913	0.913	0.912	0.913	50.07%	203.87
AFMO-BERT(Mean8)	0.909	0.909	0.909	0.909	97.96%	70.21

**Fig. 3.** Changes in the current accuracy and the best accuracy during the algorithm process as temperature decreases.**Fig. 4.** Changes in the number of selected dimensions and the best accuracy as the temperature decreases.

extraction cost of large pretrained models such as BERT, this runtime isolates the cost of the proposed optimization mechanism itself. Table 5 displays the total optimization time over 100 runs with independent random seeds. As shown in Column “Runtime” in of Table 5, the total runtime ranges from 70 seconds to 204 seconds across different settings, implying a moderate average cost per run.

Fig. 3 illustrates the changes in both the current accuracy and the achieved optimal accuracy as the temperature decreases during the simulated annealing algorithm. The figure clearly demonstrates that the simulated annealing algorithm continually adjusts the accuracy and gradually improves the results compared to the initial accuracy.

Fig. 4 illustrates the changes in both the optimal accuracy and the number of selected dimensions at each point during the simulated annealing algorithm as the temperature decreases. The figure clearly demonstrates that as the simulated annealing algorithm progresses and the temperature decreases, the optimal accuracy achieved increases while the number of selected dimensions decreases.

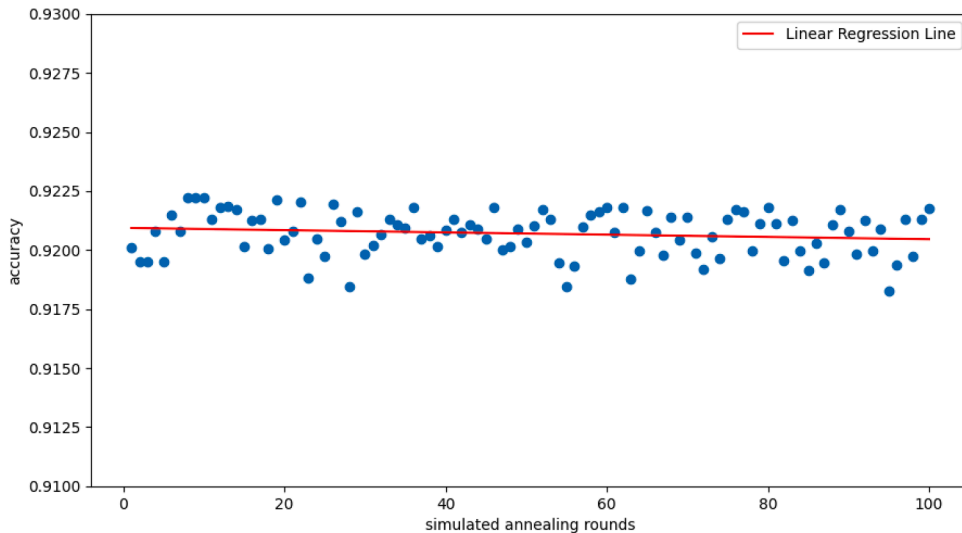


Fig. 5. The results obtained from 100 independent runs of AFMO.

Fig. 5 reports the accuracy obtained from 100 independent runs of AFMO under different random seeds. The results are tightly clustered within a narrow range (approximately 0.918–0.923), and the fitted trend line is nearly flat. This concentration suggests that the observed performance is not driven by favourable random seeds, but reflects a stable optimization behaviour.

5. Conclusion

We propose AFMO, a multimodal framework for fake news detection that aims to combine heterogeneous signals while explicitly reducing cross-modal interference. AFMO first extracts modality-specific representations from news articles using BERT for text and VGG-19 for images. To improve data reliability, it then removes anomalous training instances identified by complementary outlier detectors based on Mahalanobis distance and KNN. On the resulting curated set, AFMO applies simulated annealing to select a compact, discriminative subset of text and image dimensions for fusion, suppressing redundant or disruptive cues and yielding a higher-quality multimodal representation. The selected features are finally fed to a classifier for veracity prediction. Extensive experiments against competitive baselines demonstrate the effectiveness of AFMO for multimodal fake news detection. Future work will incorporate user-level signals (e.g., follower and friend counts) to further enhance robustness and accuracy, enabling a more comprehensive treatment of the social context in which misinformation is produced and propagated.

CRedit authorship contribution statement

Lei Wu: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization; **Guanghui Wang:** Supervision, Project administration, Funding acquisition.

Data availability

Data will be made available on request.

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